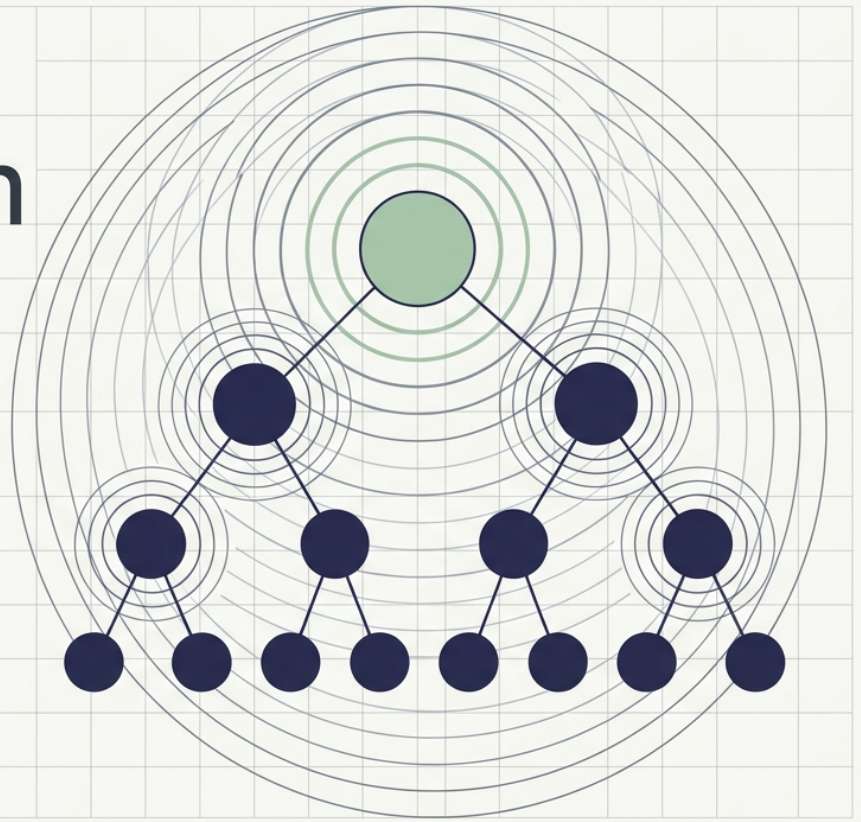


Breadth First Search (BFS)

The Coding Interview Toolkit

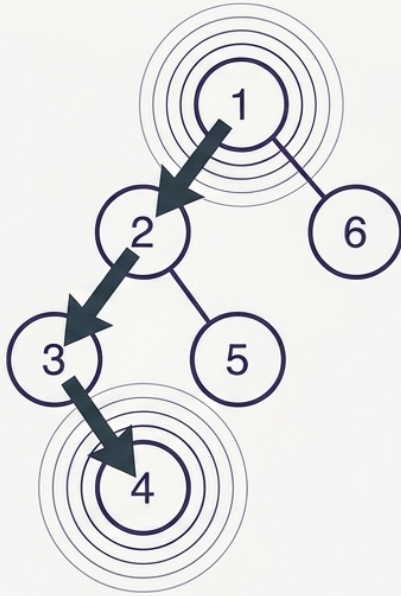
Mastering level-by-level tree traversal and queue mechanics.



TWO SPELLS, TWO SHAPES

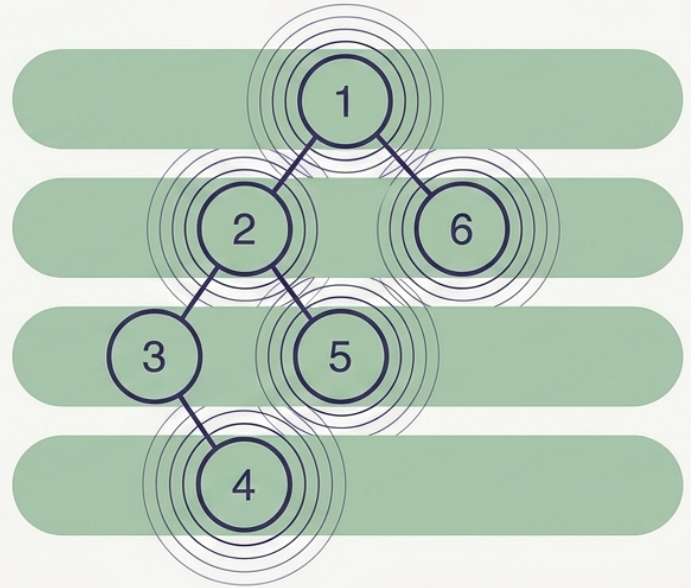
DEPTH FIRST SEARCH (DFS)

Reaches for depth (child nodes) first before breadth.



BREADTH FIRST SEARCH (BFS)

Visits all nodes in a level/depth before starting to visit the next level.



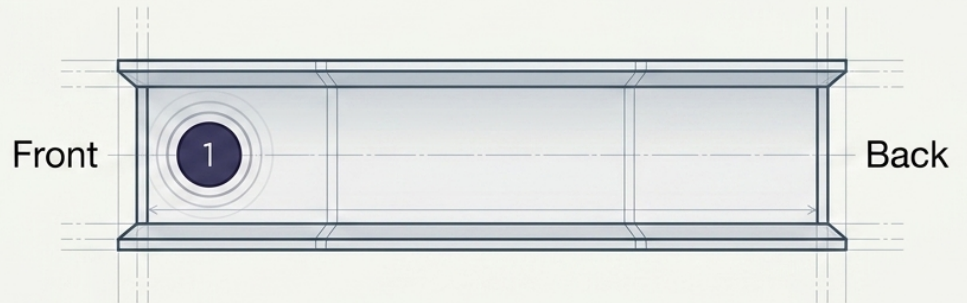
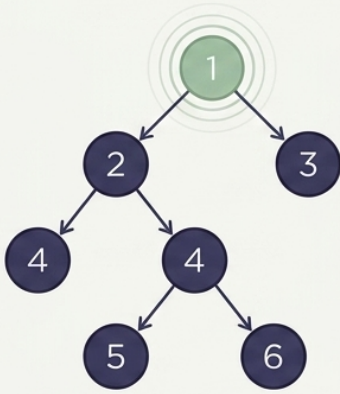
THE DECISION MATRIX: DFS VS. BFS

	DEPTH FIRST SEARCH	BREADTH FIRST SEARCH
THE TARGET	Finding nodes far away from the root. 	Finding nodes close or closest to the root. 
UNDER THE HOOD	Uses Recursion / The Call Stack (Last In, First Out). 	Uses a Queue (First In, First Out). 

TAKEAWAY: Since both traverse tree nodes, the only differences are caused by their traversal order.

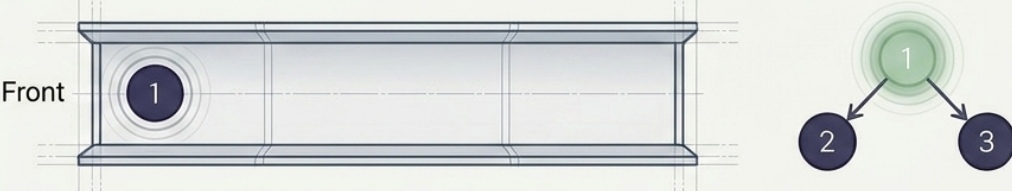
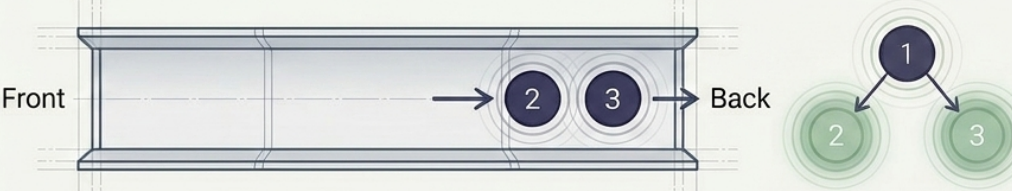
THE MECHANICS: THE FIFO CONVEYOR BELT

While DFS uses a stack, BFS uses a queue (First In, First Out).



The Golden Rule: When we dequeue a node, we enqueue its children.

TRACING THE BFS ENGINE

Step 1: Start with Node 1 in the queue.	 The diagram shows a queue represented as a horizontal container with three compartments. The first compartment contains a dark blue circle with the number 1. An arrow labeled 'Front' points to this compartment. To the right, a tree structure shows a root node 1 (dark blue) with two children, nodes 2 and 3 (dark blue).
Step 2: Pop from the Front (Dequeue).	 The diagram shows the same queue. Node 1 is still in the first compartment, but two arrows point away from it, one to the left and one to the right, indicating it is being removed. The 'Front' label is still present.
Step 3: Push children to the Back (Enqueue).	 The diagram shows the queue with two compartments. The first compartment is empty, and the second compartment contains two dark blue circles, 2 and 3. Arrows point from the left and right towards these nodes. The 'Front' label is on the left, and the 'Back' label is on the right. To the right, a tree structure shows a root node 1 (dark blue) with two children, nodes 2 and 3 (green).

THE PYTHON BLUEPRINT

```
from collections import deque
```

```
def bfs_by_queue(root):
```

```
    queue = deque([root])
```

```
    while len(queue) > 0:
```

```
        node = queue.popleft()
```

```
        if OK(node):
```

```
            return FOUND(node)
```

```
            for child in node.children:
```

```
                queue.append(child)
```

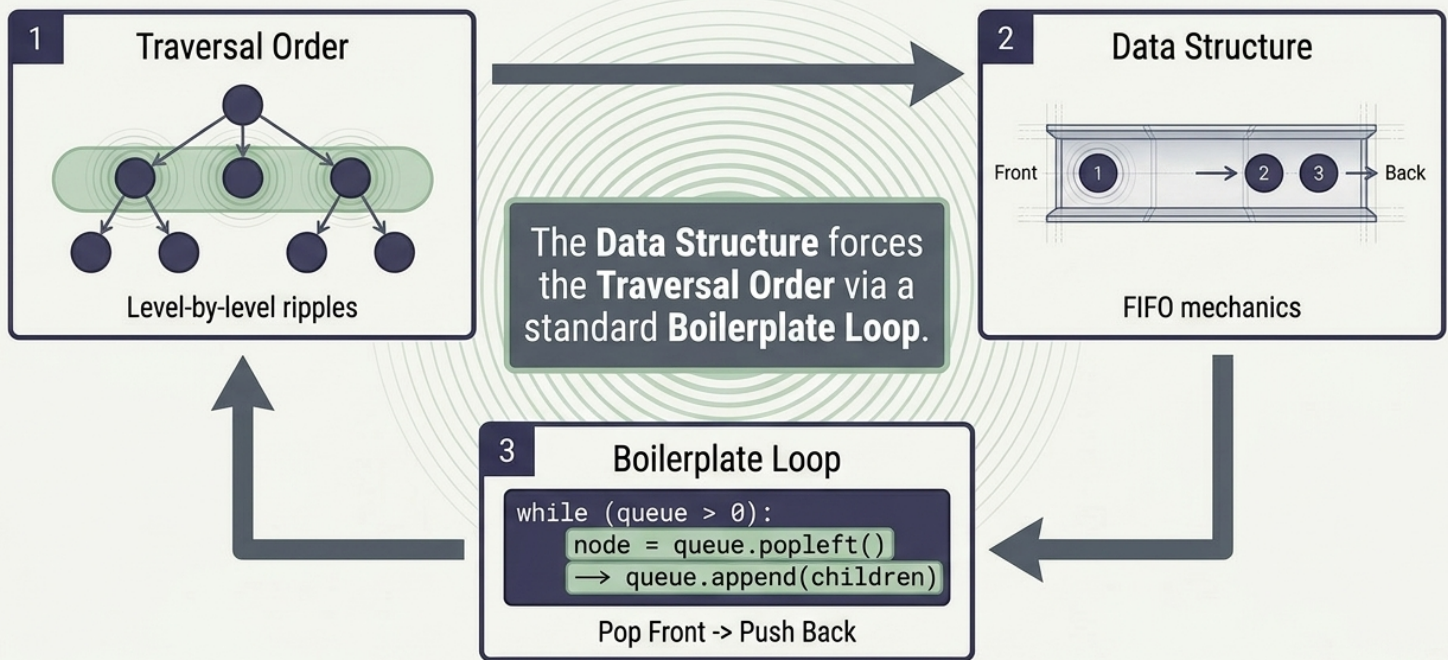
```
    return NOT_FOUND
```

The Kickstart: Need at least one element to kick start the process.

The Engine: As long as there is an element, dequeue from the front.

The Expansion: Right after we dequeue an element, enqueue its children.

THE INTERVIEWER'S MENTAL MAP



Lock this triad into memory, and BFS transforms from a memorized script into an intuitive reflex.